

AWATERE DISTRICT COUNCIL

INQUIRY INTO THE RANGITOTO RESERVOIR CONTAMINATION EVENT

Report of the Inquiry — 28 February 2026 — File ref: INQ-2026-003

Executive summary

On 14 November 2025, a contamination event was detected at the Rangitoto Reservoir, resulting in a boil-water notice affecting approximately 4,200 households in the Kowhai Precinct and surrounding areas for 72 hours. This Inquiry, convened by the Chief Executive, examined the causes, identified organisational and individual failures, and recommends protective measures. The report draws on privileged legal advice and includes candid findings on deferred maintenance practices.

Parties and persons interviewed

Inquiry Commissioner

Name: Helen Margaret Ashworth

Role: Independent Inquiry Commissioner

Date of birth: 4 August 1971

Contact: 021 908 4433, helen.ashworth@ashworthlaw.co.nz

Postal: PO Box 2184, Wellington 6011

Technical Advisor

Name: Dr Tenisha Marama Aalbers

Role: Hydrogeologist (external appointment)

Date of birth: 19 January 1982

IRD: 092-415-881

Contact: 027 552 9140, t.aalbers@aalbers-hydro.nz

Address: 47 Pohutukawa Terrace, Tauranga 3110

Senior Council Officer (subject)

Name: Gareth Alexander Thornton

Role: Group Manager — Infrastructure (until December 2025)

Date of birth: 27 November 1968

Contact: 021 744 6221, g.thornton@personalmail.example.nz

Home address: 12 Miro Street, Awatere 4310

Parties — continued

Contractor representative

Name: Ms Rua Maia Henderson

Role: Site Supervisor, Pacific Flowline Services

Date of birth: 11 May 1978

Contact: 022 803 4416, r.henderson@pacificflowline.example.co.nz

Postal: Level 3 / 188 Vivian Street, New Plymouth 4310

Witness (maintenance technician)

Name: Mr Ieremia Hemi Valeafou

Role: Maintenance Technician, Water Treatment

Date of birth: 2 October 1988

IRD: 115-328-904

Contact: 027 220 9915, ivaleafou@example.nz

Home address: 9 Rimu Place, Awatere 4310

Background

The Rangitoto Reservoir services the Kowhai Precinct and adjacent suburbs, supplying treated water to approximately 4,200 residential connections and twelve commercial premises. The plant comprises a sedimentation basin, dual-stage chlorination system, and two reservoirs. Chlorine dosing is managed by a programmable logic controller with automated feedback from in-line residual sensors, supplemented by manual-mode operation when the automated system is offline.

Routine maintenance of the chlorine dosing equipment is undertaken under a maintenance contract with Pacific Flowline Services. Under that contract, Pacific Flowline attend quarterly for scheduled servicing. In-between quarters, Council's own Maintenance Technicians are responsible for daily inspection, sensor calibration, and stockpile management. Mr Valeafou was the rostered technician on the days leading up to the event and was the first to detect the residual-chlorine deficit that triggered the contamination response.

The Inquiry interviewed all named parties between 12 January 2026 and 17 February 2026. Transcripts are appended. References to individuals in this report use full names on first mention and honorific-plus-surname form thereafter, except where direct quotation requires otherwise.

Timeline of events

11 November 2025 (D-3): Pacific Flowline attended for scheduled quarterly servicing. Ms Henderson's field report, reviewed by the Inquiry, records that the primary dosing pump was serviced and that the secondary backup pump was found to have a calibration drift of 14% beyond tolerance. Ms Henderson's report recommends the secondary pump be taken offline for workshop service, leaving the primary pump as sole active unit until a replacement part arrives.

12 November 2025 (D-2): Mr Thornton, as Group Manager, was copied on Ms Henderson's recommendation via email at 14:22 NZDT. No response is recorded in the Council email archive. Mr Valeafou, on shift that afternoon, noted in the plant logbook that the secondary backup pump was flagged out of service and that the primary pump was operating within tolerance.

13 November 2025 (D-1): Mr Valeafou's morning inspection noted residual-chlorine readings 0.3 mg/L below the normal operating band but within the statutory minimum. The logbook records that he attempted to reach Mr Thornton's office to escalate the drift but was informed Mr Thornton was off-site for a budget session. The acting Group Manager's diary does not show any callback.

14 November 2025, 06:40 (D-day): Mr Valeafou's pre-shift sensor sweep detected residual chlorine at 0.09 mg/L — below the statutory minimum of 0.20 mg/L. He immediately triggered the plant's automated boil-water protocol and contacted Ms Henderson, who arrived on site at 07:15.

14 November 2025, 08:30: Council's Emergency Management Coordinator issued a boil-water notice to the 4,200 affected households via text-message alert and radio. Dr Aalbers was contacted at 09:10 and retained as independent technical advisor the same afternoon.

14-16 November 2025: The primary dosing pump was found to have a cracked diaphragm. Pacific Flowline engineers replaced the diaphragm under emergency call-out. Dr Aalbers recommended a parallel sampling programme across all service reticulation zones to confirm restoration; this programme was completed on 17 November.

17 November 2025, 18:00: Boil-water notice lifted after 72 hours following clearance by Dr Aalbers and the Ministry of Health regional officer.

Technical investigation — root cause

The Inquiry accepts Dr Aalbers' technical findings in full. Her report, attached as Appendix A, concludes that the contamination event had two proximate causes:

(1) The primary dosing pump's diaphragm failed over a period of 8 to 14 hours beginning during the afternoon of 13 November 2025. The diaphragm had passed its last scheduled replacement date (May 2025) by six months. Dr Aalbers notes that diaphragm replacements are scheduled at six-monthly intervals in the manufacturer's specification and that the May 2025 replacement was the last recorded replacement in the plant maintenance log.

(2) The secondary backup pump had been out of service since 11 November 2025 pending the arrival of a calibration-controller replacement ordered by Ms Henderson on the same date. In Dr Aalbers' assessment, had the secondary backup been operational, automatic failover would have maintained the residual-chlorine band within the statutory minimum and no boil-water notice would have been necessary.

Dr Aalbers further identifies three contributing systemic factors, each of which is discussed at greater length in the Findings section:

(a) The plant's automated alerting did not distinguish between the statutory minimum (0.20 mg/L) and the internal operating band lower bound (0.50 mg/L). Dr Aalbers states that the operating-band lower bound is the threshold at which backup chlorination should automatically engage, and that relying on the statutory minimum alone leaves an insufficient buffer for the 8-14 hour decay window observed on 13-14 November.

(b) The maintenance log does not flag elapsed-time counters for diaphragm components. Dr Aalbers recommends a mandatory vulnerability assessment for chlorine dosing equipment on a six-monthly cycle, with explicit flagging of components approaching replacement due date.

(c) The contamination response plan activation threshold of 4.5 mg/L (for the inverse event — over-chlorination) had been tabletopped in 2024 but the under-dosing threshold had not been rehearsed since 2019. Dr Aalbers considers this a material gap in emergency preparedness.

Witness testimony — Dr Aalbers

Dr Aalbers was interviewed on 22 January 2026. She confirmed the findings set out in her technical report and was asked to elaborate on matters the Inquiry considered material to organisational accountability.

On the question of whether Mr Thornton should have acted on Ms Henderson's 11 November warning, Dr Aalbers stated: "A reasonable infrastructure manager, on receipt of a field report flagging a backup system out of tolerance, would either action same-day replacement or escalate the risk tolerance decision to the Chief Executive. Deferring the decision for 48 hours in plain hope that the primary unit would hold was, in my view, not within the band of reasonable technical judgement."

Asked whether the 2024 risk register had identified this specific failure mode, Dr Aalbers confirmed that the 2024 register had recorded "concurrent failure of primary and backup chlorine dosing" as a high-likelihood / high-consequence risk, rated Priority 1 for mitigation. Dr Aalbers confirmed that the mitigation recommendation in the 2024 register was identical to her recommendation in the present Inquiry: namely, maintain backup-pump redundancy at all times and implement mandatory vulnerability assessment protocols for chlorine dosing equipment on a six-monthly cycle.

Asked whether Council's public framing of the 2024 audit as "clean" was accurate, Dr Aalbers declined to offer a regulatory opinion but noted that the audit's detailed findings were not consistent with the single-sentence public summary circulated at the time. She referred the Inquiry to Appendix B of her report, which reproduces the relevant audit findings verbatim.

Dr Aalbers' view, offered in candid form and recorded with her permission: "The 2024 audit was not a clean audit. It was an audit with one material adverse finding and the adverse finding was the one that materialised fourteen months later. That is a plain statement of fact rather than a judgement, but I accept it is a statement the Council may not welcome."

Aalbers' further remarks on the contamination response plan and the adequacy of the public-notification protocol are set out in the Findings section.

Witness testimony — Mr Thornton

Mr Thornton was interviewed on 28 January 2026. The Inquiry gave Mr Thornton advance notice of the matters to be covered and the opportunity to respond to Ms Henderson's 11 November email in writing prior to interview. Mr Thornton provided a written response on 26 January 2026 (Appendix C) and was interviewed in person on the scheduled date.

Mr Thornton confirmed receipt of Ms Henderson's 11 November 14:22 email. He accepted that he had not responded to the email on 12 November and had not personally actioned the backup-pump replacement recommendation. Mr Thornton's explanation, as he put it: "I was in the final week of the draft budget cycle and I will be candid — I prioritised the budget submission over a maintenance escalation that, on the face of the email, did not read to me as same-day urgent."

Asked whether he had seen the 2024 risk register's Priority 1 entry on concurrent dosing-pump failure, Mr Thornton stated that he had reviewed the register at the time of its publication in March 2024 and had noted the Priority 1 entry. He stated that he had relied on existing redundancy — the two-pump architecture — to meet the residual risk and that he had not directed any further mitigation activity beyond the existing quarterly-service schedule with Pacific Flowline.

Asked whether he had personally approved the public framing of the 2024 audit as "clean", Mr Thornton confirmed that he had reviewed and approved the single-sentence public summary in March 2024. His written submission to the Inquiry acknowledges, in direct language: "In candid retrospect, the Council's organisational tolerance for maintenance deferral has drifted materially from safe practice. I was part of that drift and I bear a portion of the accountability for it."

Mr Thornton's employment with Council ended on 12 December 2025 by mutual agreement. The Inquiry was not asked to opine on the terms of that agreement and does not do so.

The Inquiry accepts Thornton's written and oral evidence as frank and finds it has been helpful in identifying the sequence of decisions that led to the 14 November event. It does not find, on balance, that Thornton's conduct was wilful, but it does find that the conduct fell short of the standard reasonably expected of a Group Manager with direct accountability for a Priority 1 public-health risk.

Witness testimony — Ms Henderson

Ms Henderson was interviewed on 4 February 2026 at Pacific Flowline's New Plymouth office. Ms Henderson confirmed the sequence of events as set out in her 11 November field report and the subsequent 14 November call-out. She stated that Pacific Flowline's standard practice, on identifying a backup pump out of tolerance, is to escalate in writing within the same shift and to request written acknowledgement within 24 hours. Her 11 November email to Mr Thornton is consistent with that practice.

Asked whether Pacific Flowline had any commercial interest in the outcome of the 14 November event, Ms Henderson confirmed that Pacific Flowline's contract includes a variation clause for emergency call-out. The variation fee claimed by Pacific Flowline for the 14-16 November diaphragm replacement was \$18,400, of which \$6,200 covered parts (at cost) and \$12,200 covered labour (at the contract's standard emergency rate). The variation has been paid in full by Council and is not in dispute.

Pacific Flowline's broader commercial arrangements with Council are governed by the maintenance contract dated 3 April 2024, which includes a commercial-in-confidence schedule detailing proprietary chlorine-dosing calibration procedures and a trade-secret schedule of Pacific Flowline's preferred parts suppliers. Henderson asked that the Inquiry respect the confidentiality undertaking in the contract. The Inquiry has done so and the commercial schedules are not reproduced in this report.

Witness testimony — Mr Valeafou

Mr Valeafou was interviewed on 6 February 2026. His evidence was consistent with the plant logbook and the automated telemetry. Mr Valeafou detailed his attempts to escalate the residual-chlorine drift on 13 November and the difficulty he had reaching Mr Thornton during the budget-session period.

Asked whether he had felt any pressure not to escalate, Mr Valeafou stated: "I did not feel pressured not to escalate. I did feel that the escalation path was not responsive on that particular day. I raised the drift with the acting Group Manager's office and I left a message. I did not get a callback." Valeafou confirmed that on 14 November he followed plant protocol exactly and contacted Henderson immediately on detecting the 0.09 mg/L reading.

The Inquiry commends Valeafou's conduct on 13-14 November and finds no fault with his actions. His logbook entries are contemporaneous and detailed; the Inquiry has relied on them extensively.

Commercial arrangements

The Inquiry reviewed Council's commercial arrangements with Pacific Flowline Services. The maintenance contract dated 3 April 2024 is a three-year term at an annual fee of \$840,000, with a commercial-in-confidence schedule detailing proprietary chlorine-dosing calibration procedures that Pacific Flowline treat as trade-secret information. The variation fee paid in respect of the 14-16 November emergency response was \$18,400 and is on the record.

Pacific Flowline also tendered for a subsequent variation, raised by Ms Henderson at the Inquiry, covering redesign of the automated alerting thresholds. That variation was quoted at \$41,500 and included a proprietary signal-processing methodology which Pacific Flowline represents as a trade secret under their commercial-in-confidence submission. The variation has not yet been accepted by Council pending the recommendations of this Inquiry.

Privileged legal advice

The Inquiry has been provided with privileged advice from Farrow & Slate Lawyers on Council's exposure under the Health Act 1956 and the Local Government Act 2002. That advice, which is attached as Appendix D and which remains subject to solicitor-client privilege, is that Council's exposure is limited but non-trivial. Counsel's settlement range, should any affected party pursue a claim, is \$180,000 to \$450,000 inclusive of legal costs.

Farrow & Slate further advise that the privileged opinion must not be released under LGOIMA without express waiver from the Chief Executive in consultation with the Mayor and the Audit and Risk Committee Chair. The Inquiry accepts that advice and has treated the privileged opinion as confidential for the purpose of this report.

The Inquiry notes that the privileged advice canvasses the question of whether Mr Thornton's conduct gives rise to any separate personal liability. Farrow & Slate conclude that it does not. The Inquiry does not disturb that conclusion.

Findings — free and frank

The Inquiry's findings below are expressed in candid form and are intended as advice to the Chief Executive and elected members. They are not formal regulatory findings and are not offered in terms of any statutory test.

Finding 1: In candid retrospect, staff concerns raised in the 2024 risk register were not actioned with appropriate urgency. The Priority 1 entry on concurrent chlorine-dosing failure was visible to three successive Group Managers and was not acted on until the event it predicted had occurred. That is a plain failure of risk-management process, not a failure of any individual officer's good faith.

Finding 2: The Inquiry finds, frankly, that the prior Group Manager's public framing of the 2024 audit as "clean" was inconsistent with the audit's actual findings. The audit was not clean. It contained one material adverse finding, and that finding was the proximate cause of the November 2025 event. Misrepresenting the audit to the public and to elected members denied those audiences the information they needed to hold officers to account.

Finding 3: The organisational tolerance for maintenance deferral has, in the Inquiry's view, drifted materially from safe practice. Officers routinely deferred six-monthly replacements to nine- or twelve-month cycles on the implicit assumption that redundancy would absorb the risk. That assumption was wrong on the facts; it was also wrong in principle. Redundancy is a margin for unexpected failure, not a licence for expected failure.

Finding 4: The escalation path on 12-13 November was not functional. Mr Valeafou's attempts to reach Mr Thornton on 13 November were met with a budget-session absence and an unresponsive acting Group Manager's office. The Inquiry finds that single-point-of-failure escalation paths are incompatible with Priority 1 public-health risk management and must be replaced with a roster-backed on-call system. These findings are offered in advice rather than as judgement. The Inquiry accepts that each individual officer acted in good faith. The findings go to organisational process and culture rather than to individual conduct, except to the extent noted in respect of Mr Thornton above.

Health-safety protective measures

The Inquiry recommends the following protective measures, each of which is designed to reduce the likelihood or consequence of a similar event. Certain operational details in these measures are withheld from public release under s7(2)(d) because disclosure would reveal specific vulnerabilities that an adversary could exploit to trigger a contamination event.

Protective measure A: Mandatory six-monthly vulnerability assessment protocols for chlorine dosing equipment, with explicit component-lifecycle flagging and an escalation path that does not rely on a single officer. The detailed protocol is set out in Appendix E of this report. Appendix E contains operational information whose disclosure could endanger public safety if exploited and must not be released under LGOIMA.

Protective measure B: Emergency backup chlorination procedure at the 4.5 mg/L threshold, rehearsed at annual frequency and tested against both under-dosing and over-dosing scenarios. The under-dosing scenario has not been rehearsed since 2019 and the Inquiry considers this a material gap. The protective-measure specification includes the dispatch sequencing and alternate dosing-source coordinates for the backup supply, both of which are withheld from public release on the same s7(2)(d) basis.

Protective measure C: Contamination response plan activation thresholds, updated to include automated triggering at the internal operating-band lower bound (0.50 mg/L) rather than the statutory minimum alone, and extended to cover a stress-tested boil-water notice distribution list.

Recommendations

The Inquiry recommends: (1) adoption of protective measures A, B and C as set out above, with implementation led by the incoming Group Manager Infrastructure and reported to the Audit and Risk Committee quarterly; (2) roster-backed 24-hour on-call escalation for Priority 1 public-health risks; (3) a formal externally-facilitated review of the 2024 risk register; (4) public disclosure of this report in its non-withheld form within 10 working days of adoption by Council.

Signed:

Helen Margaret Ashworth

Independent Inquiry Commissioner

28 February 2026